

An exact-continuum analytical reference for DEM contact tests: closed-form contact increments and high-order orientation integration

Notes accompanying `scripts.PY/A_analytical_motion.py`

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Abstract

The interaction-test suite compares a discrete-element (DEM) contact implementation against an *analytical* reference trajectory. For the comparison to measure the DEM error, the reference itself must be as close to the exact continuous solution as possible. Two quantities drive everything downstream: the per-step contact *increments* ($d\theta_t$, $d\mathbf{u}_r$, $d\mathbf{u}_s$, $d\boldsymbol{\theta}_b$) consumed by the constitutive laws, and the particle *orientations* q_i, q_j . This note derives, from first principles, (i) the closed-form evaluation of every increment as an exact integral over a rotating contact frame, and (ii) a fourth-order Magnus integrator for the orientation quaternion. We explain why the previous “analytical-magnitude $\times$ end-of-step direction” construction was only first-order accurate whenever the contact frame rotates, and why the schemes here remove that error.

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1 Setting and notation

Two spheres i, j are driven through a prescribed relative motion built from a rigidly rotating contact frame plus three superimposed relative modes (twist, roll, shear). The frame rotates at

the constant angular velocity $\boldsymbol{\omega}_f$, so any material direction fixed in the frame is carried by the rotation operator

$$\mathbf{R}_b(t) = \exp([\boldsymbol{\omega}_f]_{\times} t), \quad [\boldsymbol{\omega}_f]_{\times} = \begin{pmatrix} 0 & -\omega_{f,z} & \omega_{f,y} \\ \omega_{f,z} & 0 & -\omega_{f,x} \\ -\omega_{f,y} & \omega_{f,x} & 0 \end{pmatrix}, \quad (1)$$

a proper rotation about the unit axis $\hat{\mathbf{e}} = \boldsymbol{\omega}_f / \varphi_f$ through the angle $\varphi_f t$, where $\varphi_f = \|\boldsymbol{\omega}_f\|$. Let $\mathbf{n}_0$ be the initial contact normal (unit branch direction) and $\mathbf{n}_{r0}, \mathbf{n}_{s0}$ the roll and shear axes (orthogonal to $\mathbf{n}_0$). Their instantaneous values are

$$\mathbf{n}(t) = \mathbf{R}_b(t) \mathbf{n}_0, \quad \mathbf{n}_r(t) = \mathbf{R}_b(t) \mathbf{n}_{r0}, \quad \mathbf{n}_s(t) = \mathbf{R}_b(t) \mathbf{n}_{s0}. \quad (2)$$

Every mode is modulated by a rate function of the same exponential–sinusoidal family,

$$s(\tau) = A - B \sin(w\tau + \psi) e^{k\tau}, \quad (3)$$

with mode-specific parameters $(A_X, B_X, w_X, \psi_X, k_X)$ for $X \in \{t, r, s\}$ (twist, roll, shear). The relative twist angular velocity, and the roll and shear sliding velocities of the contact point, are

$$\boldsymbol{\omega}_t(\tau) = s_t(\tau) \mathbf{n}(\tau), \quad (4)$$

$$\mathbf{v}_r(\tau) = s_r(\tau) (\mathbf{n}_r(\tau) \times \mathbf{n}(\tau)), \quad (5)$$

$$\mathbf{v}_s(\tau) = s_s(\tau) (\mathbf{n}_s(\tau) \times \mathbf{n}(\tau)), \quad (6)$$

and the per-particle angular velocities are

$$\boldsymbol{\omega}_i(\tau) = \boldsymbol{\omega}_f + \frac{1}{2} s_t \mathbf{n} + \frac{1}{2R_i} s_r \mathbf{n}_r + \frac{1}{2R_i} s_s \mathbf{n}_s, \quad \boldsymbol{\omega}_j(\tau) = \boldsymbol{\omega}_f - \frac{1}{2} s_t \mathbf{n} - \frac{1}{2R_j} s_r \mathbf{n}_r + \frac{1}{2R_j} s_s \mathbf{n}_s. \quad (7)$$

The contact laws integrate these rates over each time step $[t_{k-1}, t_k]$ of length Δt . We must evaluate, exactly,

$$d\boldsymbol{\theta}_t = \int_{t_{k-1}}^{t_k} \boldsymbol{\omega}_t d\tau, \quad d\mathbf{u}_r = \int_{t_{k-1}}^{t_k} \mathbf{v}_r d\tau, \quad d\mathbf{u}_s = \int_{t_{k-1}}^{t_k} \mathbf{v}_s d\tau, \quad d\boldsymbol{\theta}_i = \int_{t_{k-1}}^{t_k} \boldsymbol{\omega}_i d\tau, \quad (8)$$

and likewise the bending increment $d\boldsymbol{\theta}_b = d\boldsymbol{\theta}_i - d\boldsymbol{\theta}_j - d\boldsymbol{\theta}_t$. The orientations follow from integrating the attitude equation $\dot{\mathbf{R}}_i = [\boldsymbol{\omega}_i]_{\times} \mathbf{R}_i$ (Section 4).

2 Why the naive increment is only first order

Each integrand in (8) is a scalar rate times a *rotating* unit vector. The previous implementation integrated the scalar exactly but froze the direction at the end of the step. For the shear increment, for example,

$$d\mathbf{u}_s^{\text{endpoint}} = \underbrace{\left(\int_{t_{k-1}}^{t_k} s_s d\tau \right)}_{= M, \text{ exact}} (\mathbf{n}_s(t_k) \times \mathbf{n}(t_k)) = M \mathbf{R}_b(t_k) \mathbf{d}, \quad \mathbf{d} \equiv \mathbf{n}_{s0} \times \mathbf{n}_0. \quad (9)$$

The exact value is $d\mathbf{u}_s = \int s_s(\tau) \mathbf{R}_b(\tau) \mathbf{d} d\tau$, so the error of the endpoint rule is

$$d\mathbf{u}_s^{\text{endpoint}} - d\mathbf{u}_s = \int_{t_{k-1}}^{t_k} s_s(\tau) [\mathbf{R}_b(t_k) - \mathbf{R}_b(\tau)] \mathbf{d} d\tau. \quad (10)$$

Because $\mathbf{R}_b(t_k) - \mathbf{R}_b(\tau) = [\boldsymbol{\omega}_f]_{\times} \mathbf{R}_b(\tau) (t_k - \tau) + \mathcal{O}((t_k - \tau)^2)$, the bracket is $\mathcal{O}(\varphi_f \Delta t)$ across the step, and the integral is $\mathcal{O}(\varphi_f \Delta t \cdot M) = \mathcal{O}(\varphi_f \Delta t^2)$ *per step*. Summed over the $\mathcal{O}(1/\Delta t)$ steps

of a simulation this is a *global* error of $\mathcal{O}(\Delta t)$: the endpoint rule is only **first-order accurate** whenever the frame rotates ($\boldsymbol{\omega}_f \neq \mathbf{0}$). The exactness of the scalar integral M is a red herring – the frozen direction sets the order. (A midpoint evaluation $s_s(t_m) \Delta t \mathbf{R}_b(t_m) \mathbf{d}$ would be $\mathcal{O}(\Delta t^2)$ globally, better but still not exact.)

The remedy is to integrate the rotating direction *analytically* as well. As shown next, this is possible in closed form for the entire rate family (3), so the increment error is eliminated to machine precision.

3 Exact closed-form increments

3.1 One canonical integral

A single structure underlies every increment in (8). Because a proper rotation preserves cross products, $(\mathbf{R}_b \mathbf{a}) \times (\mathbf{R}_b \mathbf{b}) = \mathbf{R}_b(\mathbf{a} \times \mathbf{b})$, the roll/shear velocities reduce to one rotating vector,

$$\mathbf{v}_s(\tau) = s_s(\tau) \mathbf{R}_b(\tau) (\mathbf{n}_{s0} \times \mathbf{n}_0), \quad \mathbf{v}_r(\tau) = s_r(\tau) \mathbf{R}_b(\tau) (\mathbf{n}_{r0} \times \mathbf{n}_0), \quad (11)$$

exactly the form of the twist rate (4). Hence every increment is an instance of

$$\boxed{\mathbf{J}(\mathbf{d}; s) = \int_{t_a}^{t_b} s(\tau) \mathbf{R}_b(\tau) \mathbf{d} \, d\tau} \quad (12)$$

for a fixed seed $\mathbf{d}$ and rate s :

$$d\boldsymbol{\theta}_t = \mathbf{J}(\mathbf{n}_0; s_t), \quad d\mathbf{u}_r = \mathbf{J}(\mathbf{n}_{r0} \times \mathbf{n}_0; s_r), \quad d\mathbf{u}_s = \mathbf{J}(\mathbf{n}_{s0} \times \mathbf{n}_0; s_s), \quad (13)$$

$$d\boldsymbol{\theta}_i = \boldsymbol{\omega}_f \Delta t + \frac{1}{2} \mathbf{J}(\mathbf{n}_0; s_t) + \frac{1}{2R_i} \mathbf{J}(\mathbf{n}_{r0}; s_r) + \frac{1}{2R_i} \mathbf{J}(\mathbf{n}_{s0}; s_s). \quad (14)$$

Note the subtlety that distinguishes a *displacement* from a *rotation rate*: the roll/shear displacements seed $\mathbf{J}$ with the cross product $\mathbf{n}_{X0} \times \mathbf{n}_0$ (their velocity is a moment), while the per-particle rotation rates seed it with the bare direction $\mathbf{n}_{X0}$.

3.2 Rodrigues decomposition of the rotating seed

Rodrigues' formula resolves $\mathbf{R}_b(\tau) \mathbf{d}$ into a part fixed along the axis and a part that circulates:

$$\mathbf{R}_b(\tau) \mathbf{d} = \mathbf{d} \cos(\varphi_f \tau) + (\hat{\mathbf{e}} \times \mathbf{d}) \sin(\varphi_f \tau) + \hat{\mathbf{e}}(\hat{\mathbf{e}} \cdot \mathbf{d})(1 - \cos(\varphi_f \tau)) = \mathbf{a} \cos(\varphi_f \tau) + \mathbf{b} \sin(\varphi_f \tau) + \mathbf{c}, \quad (15)$$

with the three constant vectors

$$\mathbf{c} = \hat{\mathbf{e}}(\hat{\mathbf{e}} \cdot \mathbf{d}) \quad (\text{axial}), \quad \mathbf{a} = \mathbf{d} - \mathbf{c} \quad (\perp, \text{cos-part}), \quad \mathbf{b} = \hat{\mathbf{e}} \times \mathbf{d} \quad (\perp, \text{sin-part}). \quad (16)$$

Substituting into (12) factors the vector and scalar parts completely:

$$\begin{aligned} M &= \int_{t_a}^{t_b} s(\tau) \, d\tau, \\ \mathbf{J}(\mathbf{d}; s) &= \mathbf{a} C + \mathbf{b} S + \mathbf{c} M, \quad C = \int_{t_a}^{t_b} s(\tau) \cos(\varphi_f \tau) \, d\tau, \\ S &= \int_{t_a}^{t_b} s(\tau) \sin(\varphi_f \tau) \, d\tau. \end{aligned} \quad (17)$$

We have reduced a rotating vector integral to three *scalar* integrals.

3.3 The three scalar integrals

With s from (3), all three reduce to the two elementary antiderivatives

$$E_s(k, \Omega, \psi; \tau) \equiv \int e^{k\tau} \sin(\Omega\tau + \psi) d\tau = \frac{e^{k\tau} (k \sin(\Omega\tau + \psi) - \Omega \cos(\Omega\tau + \psi))}{k^2 + \Omega^2}, \quad (18)$$

$$E_c(k, \Omega, \psi; \tau) \equiv \int e^{k\tau} \cos(\Omega\tau + \psi) d\tau = \frac{e^{k\tau} (k \cos(\Omega\tau + \psi) + \Omega \sin(\Omega\tau + \psi))}{k^2 + \Omega^2}, \quad (19)$$

(valid for $k^2 + \Omega^2 \neq 0$; for $k = \Omega = 0$ the integrand is the constant $\sin \psi$ or $\cos \psi$). Using the product-to-sum identities $\sin \alpha \cos \beta = \frac{1}{2}[\sin(\alpha + \beta) + \sin(\alpha - \beta)]$ and $\sin \alpha \sin \beta = \frac{1}{2}[\cos(\alpha - \beta) - \cos(\alpha + \beta)]$ with $\alpha = w\tau + \psi$, $\beta = \varphi_f \tau$, the integrals evaluate (as $F|_{t_a}^{t_b} = F(t_b) - F(t_a)$) to

$$M = A\tau - B E_s(k, w, \psi; \tau) \Big|_{t_a}^{t_b}, \quad (20)$$

$$C = \left[A \frac{\sin(\varphi_f \tau)}{\varphi_f} - \frac{B}{2} (E_s(k, w + \varphi_f, \psi; \tau) + E_s(k, w - \varphi_f, \psi; \tau)) \right] \Big|_{t_a}^{t_b}, \quad (21)$$

$$S = \left[-A \frac{\cos(\varphi_f \tau)}{\varphi_f} - \frac{B}{2} (E_c(k, w - \varphi_f, \psi; \tau) - E_c(k, w + \varphi_f, \psi; \tau)) \right] \Big|_{t_a}^{t_b}. \quad (22)$$

Equations (12)–(22) are the exact increment, evaluated with a handful of exp, sin, cos calls and no time stepping.

3.4 The non-rotating limit

As $\varphi_f \rightarrow 0$ the frame stops and $\mathbf{R}_b(\tau) \rightarrow \mathbf{I}$, so $\mathbf{J}(\mathbf{d}; s) \rightarrow (\int s d\tau) \mathbf{d} = M \mathbf{d}$. This is exactly the old “scalar integral $\times$ constant direction” formula: **when $\omega_f = 0$ the directions never move and the previous construction was already exact**; the error identified in Section 2 appears only for a rotating frame. The code takes this $M \mathbf{d}$ branch directly when $\varphi_f = 0$, avoiding the undefined axis $\hat{\mathbf{e}}$.

4 Orientation: a fourth-order Magnus integrator

4.1 Why composing $\int \omega d\tau$ is not exact

A particle’s orientation obeys the linear but *non-autonomous* attitude equation

$$\dot{\mathbf{R}}(t) = [\boldsymbol{\omega}(t)]_{\times} \mathbf{R}(t), \quad \text{equivalently} \quad \dot{q} = \frac{1}{2} \boldsymbol{\omega}(t) \otimes q, \quad (23)$$

with $\boldsymbol{\omega}$ in the world frame. Over one step the exact update is $\mathbf{R}(t_k) = \exp(\boldsymbol{\Omega}) \mathbf{R}(t_{k-1})$, where the generator $\boldsymbol{\Omega}$ is given by the *Magnus expansion*

$$\boldsymbol{\Omega} = \int_{t_{k-1}}^{t_k} \boldsymbol{\omega} d\tau + \frac{1}{2} \int_{t_{k-1}}^{t_k} d\tau_1 \int_{t_{k-1}}^{\tau_1} d\tau_2 [\boldsymbol{\omega}(\tau_1), \boldsymbol{\omega}(\tau_2)] + \dots, \quad (24)$$

and the Lie bracket on $\mathfrak{so}(3)$ is the cross product, $[\mathbf{a}, \mathbf{b}] = \mathbf{a} \times \mathbf{b}$. The first term is the plain integral of the angular velocity; the second is the *coning* (non-commutativity) correction that arises because rotations about different axes do not commute.

Two errors therefore lurk. (i) If $\int \boldsymbol{\omega} d\tau$ is itself built from end-of-step directions it is only first order, exactly as in Section 2. (ii) Even with the integral done exactly, simply exponentiating it drops the commutator term, leaving a per-step error of $\mathcal{O}(\Delta t^3)$, i.e. *second order* globally. Both must be addressed to obtain a high-fidelity reference orientation.

4.2 Two-node Gauss–Magnus (order 4)

The fourth-order Magnus method evaluates the angular velocity at the two Gauss–Legendre nodes of the step,

$$\tau_{1,2} = t_{k-1} + \left(\frac{1}{2} \mp \frac{\sqrt{3}}{6}\right)\Delta t, \quad \omega_1 = \omega(\tau_1), \quad \omega_2 = \omega(\tau_2), \quad (25)$$

and forms

$$\boxed{\Omega = \frac{1}{2}\Delta t (\omega_1 + \omega_2) + \frac{\sqrt{3}}{12}\Delta t^2 (\omega_2 \times \omega_1)}, \quad q_k = \exp(\Omega) \otimes q_{k-1}. \quad (26)$$

The first term is the two-point Gauss quadrature of $\int \omega \, d\tau$ (exact for cubics); the second is the leading coning term. The scheme is locally $\mathcal{O}(\Delta t^5)$, hence $\mathcal{O}(\Delta t^4)$ globally. Because we can evaluate $\omega_i(\tau)$ in closed form anywhere (the expressions (7)), the Gauss samples cost nothing extra in accuracy.

4.3 Derivation of the coning coefficient

To see that (26) reproduces the coning term, expand ω about the step midpoint t_m as $\omega(\tau) = \omega_m + \dot{\omega}_m(\tau - t_m) + \mathcal{O}(\Delta t^2)$. The bracket in (24) is, to leading order,

$$[\omega(\tau_1), \omega(\tau_2)] = (\omega_m + \dot{\omega}_m(\tau_1 - t_m)) \times (\omega_m + \dot{\omega}_m(\tau_2 - t_m)) = (\omega_m \times \dot{\omega}_m)(\tau_2 - \tau_1) + \mathcal{O}(\Delta t^2), \quad (27)$$

using $\omega_m \times \omega_m = \mathbf{0}$. Integrating over the triangle $t_{k-1} \leq \tau_2 \leq \tau_1 \leq t_k$ gives $\iint (\tau_2 - \tau_1) = -\Delta t^3/6$, so the exact coning contribution is

$$\frac{1}{2} \iint [\omega(\tau_1), \omega(\tau_2)] = -\frac{1}{12}\Delta t^3 (\omega_m \times \dot{\omega}_m). \quad (28)$$

For the Gauss samples, with $a = \frac{\sqrt{3}}{6}\Delta t$ one has $\omega_{1,2} = \omega_m \mp a\dot{\omega}_m$, hence $\omega_1 \times \omega_2 = 2a (\omega_m \times \dot{\omega}_m) = \frac{\sqrt{3}}{3}\Delta t (\omega_m \times \dot{\omega}_m)$. Therefore

$$\frac{\sqrt{3}}{12}\Delta t^2 (\omega_2 \times \omega_1) = -\frac{\sqrt{3}}{12}\Delta t^2 (\omega_1 \times \omega_2) = -\frac{1}{12}\Delta t^3 (\omega_m \times \dot{\omega}_m), \quad (29)$$

which matches (28) exactly. The commutator term in (26) is the coning correction, and its sign and coefficient are fixed.

4.4 Why the increments need no coning, but q does

The increments $d\theta_t, d\mathbf{u}_r, d\mathbf{u}_s, d\theta_b$ are *ordinary integrals* of rates – physical angle/length increments that the constitutive laws accumulate. They contain no commutator: coning is an artefact of *composing finite rotations*, not of the integral itself. That is why Section 3 integrates them exactly with no Magnus term, while the orientation – which *is* a finite-rotation composition – carries the coning term (26). Both are exact-continuum to the order quoted, so they remain mutually consistent (Section 5).

5 Why this improves the results

Order of accuracy. Table 1 collects the global error orders for a rotating frame ($\omega_f \neq \mathbf{0}$). The previous endpoint construction was first order; the schemes here make the increments exact (machine precision) and the orientation fourth order.

Quantity	Previous scheme	This work
Increments $d\theta_t, d\mathbf{u}_r, d\mathbf{u}_s, d\theta_b$	exact scalar $\times$ endpoint dir., $\mathcal{O}(\Delta t)$	closed form, exact ($\sim$ machine)
Orientation q_i, q_j	endpoint generator, $\mathcal{O}(\Delta t)$	Gauss–Magnus order 4, $\mathcal{O}(\Delta t^4)$

Table 1: Global accuracy for a rotating contact frame. For $\boldsymbol{\omega}_f = \mathbf{0}$ the directions are constant and the previous increments were already exact (Section 3).

Consistency of the reference. The orientations q_i, q_j are *imposed* on the DEM bodies every step, and the solver reconstructs the relative twist and bending from that imposed orientation; meanwhile the analytical reference torques are built from $d\theta_t$ and $d\theta_b$. If q and the increments came from different discrete rules they would disagree even for a perfect constitutive law. Deriving both from the same exact kinematics removes this internal inconsistency.

Isolating the DEM error. Once the reference is exact to machine precision, any residual in the DEM–vs–analytical comparison reflects the DEM scheme (its own time integration and constitutive implementation), not the reference. The test suite then measures what it is meant to measure.

Numerical confirmation. Evaluated against adaptive quadrature of the integrands in (8), the closed-form increments agree to $\sim 10^{-15}$ (round-off). The Magnus orientation agrees with a tolerance- 10^{-12} adaptive ODE solution of $\dot{q} = \frac{1}{2}\boldsymbol{\omega} \otimes q$ to $\sim 3 \times 10^{-11}$ (geodesic angle) at the production step size. By contrast the endpoint scheme exhibits the textbook $\mathcal{O}(\Delta t)$ convergence of Table 1.

6 Map to the implementation

All of the above lives in `scripts_PY/A_analytical_motion.py`:

- `_expsin_antideriv, _expcos_antideriv` – the antiderivatives E_s, E_c , including the $k = \Omega = 0$ guard.
- `_rotated_integral(d0, A,B,w,ψ,k, t_a,t_b)` – the canonical integral $\mathbf{J}$, assembling $\mathbf{a}C + \mathbf{b}S + \mathbf{c}M$ via (20)–(22), with the $\varphi_f = 0 \Rightarrow M\mathbf{d}$ fast path. It is called with the seeds of (13)–(14) to fill $d\theta_t, d\mathbf{u}_r, d\mathbf{u}_s$ and the per-particle $d\theta_i$ (hence $d\theta_b$).
- the Magnus block – evaluates $\boldsymbol{\omega}_{i,j}$ at the two Gauss nodes via `_vel_at` and forms (26); `my_integrate_rotation` composes the per-step generators into q_i, q_j .

The whole routine is vectorised over the time axis: (20)–(22) are array expressions in t_a, t_b , and the Gauss samples are batched, so the exact reference is also the fast one.